Code:

```
import pandas as pd
import matplotlib.pyplot as plt
data = {
    "Grade": ["F", "E", "D", "C", "B", "A", "S"],
    "Number_of_Students": [3, 4, 5, 8, 9, 7, 4]
}
df = pd.DataFrame(data)
df.to_csv("student_grades.csv", index=False)
df = pd.read_csv("student_grades.csv")
print("Student Grade Data:")
print(df)
expected_grades = {"F", "E", "D", "C", "B", "A", "S"}
if set(df["Grade"]) != expected_grades or len(df) != 7:
```

```
    raise ValueError("All seven grades must be present exactly
once.")
if df["Grade"].duplicated().any():
    raise ValueError("Duplicate grades are not allowed.")
if df["Number_of_Students"].isnull().any():
    raise ValueError("Student counts cannot be empty.")
if (df["Number_of_Students"] < 0).any():
    raise ValueError("Student counts cannot be negative.")
if not pd.api.types.is_integer_dtype(df["Number_of_Students"]):
    raise ValueError("Student counts must be integers.")
total = df["Number_of_Students"].sum()
if total != 40:
    raise ValueError("Total number of students must be 40.")
print("\nData validation successful!")
```

```
df["Percentage"] = df["Number_of_Students"] / total * 100
print("\nGrade-wise Student Details:")
print(df.to_string(
    index=False,
    formatters={"Percentage": "{:.2f}%".format}
))
plt.figure(figsize=(8, 6))
plt.pie(
    df["Number_of_Students"],
    labels=df["Grade"],
    autopct="%1.1f%%",
    startangle=90
)
plt.title("Percentage Distribution of Student Grades")
```

```
plt.legend(df["Grade"], title="Grades")
plt.savefig("grade_pie_chart.png", dpi=300, bbox_inches="tight")
plt.show()
plt.figure(figsize=(8, 6))
bars = plt.bar(
    df["Grade"],
    df["Number_of_Students"],
    label="Number of Students"
)
plt.xlabel("Grades")
plt.ylabel("Number of Students")
plt.title("Grade-wise Student Performance")
plt.legend()
for bar in bars:
```

```
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width() / 2,
        height,
        str(int(height)),
        ha="center",
        va="bottom"
    )
plt.savefig("grade_bar_chart.png", dpi=300, bbox_inches="tight")
plt.show()
max_row = df.loc[df["Number_of_Students"].idxmax()]
min_row = df.loc[df["Number_of_Students"].idxmin()]
print("\nHighest Grade Count:")
```

```
print(max_row["Grade"], "-", max_row["Number_of_Students"], "students")

print("\nLowest Grade Count:")

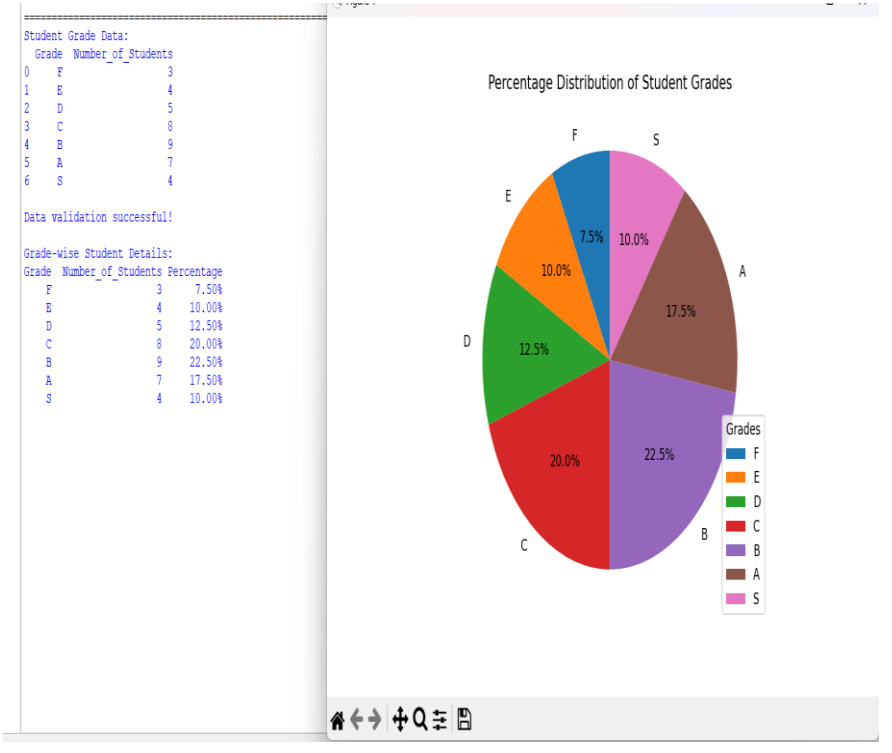
print(min_row["Grade"], "-", min_row["Number_of_Students"], "students")

print("\nCharts saved successfully:")

print("grade_pie_chart.png")

print("grade_bar_chart.png")
```

OUTPUT:



```
Student Grade Data:
  Grade  Number_of_Students
0     F                   3
1     E                   4
2     D                   5
3     C                   8
4     B                   9
5     A                   7
6     S                   4

Data validation successful!

Grade-wise Student Details:
Grade  Number_of_Students Percentage
    F                   3      7.50%
    E                   4     10.00%
    D                   5     12.50%
    C                   8     20.00%
    B                   9     22.50%
    A                   7     17.50%
    S                   4     10.00%
```